Coding & Solution

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Solution Summary:

Field	Details
Repository Link / URL	**https://github.com/Rahul-DS633/Crime-Vision-AI**
Programming Language(s)	Python 3.10 / 3.11 / 3.13, JavaScript (ES6+), HTML5, CSS3
Framework(s) Used	Flask, PyTorch, Hugging Face Transformers, OpenCV, Three.js WebGL, ReportLab Platypus
Key Features Implemented	Multimodal 3D Scene Reconstruction, OpenCV Video Evidence Engine (Vehicles, People & Objects Detection), RT-DETR Detection, Whisper Speech-to-Text with Waveform Visualizer, 4-Language Translation, 1.0Hz CCTV Timeline, Resilient Forensic AI Chatbot with Quick Chips, SHA-256 PDF Reports.
Pending Features	**None** (All core functional requirements and milestones 100% completed and verified).
Setup / Run Instructions	git clone https://github.com/Rahul-DS633/Crime-Vision-AI pip install -r backend/requirements.txt python backend/app.py (Open http://localhost:5000) python backend/test_external_data.py --all

Complete Project Code Inventory (48 Files Uploaded to Repository):

Module / File Path	Language	Primary Functionality & Implementation Highlights
backend/app.py	Python	Main Flask application entry point; registers all blueprints, CORS, static routes, lazy-loader.
backend/config.py	Python	Centralized paths (UPLOAD_DIR, REPORTS_DIR, SESSIONS_DIR), device selector, model parameters.
backend/routes/processing_bp.py	Python	Endpoints: POST /api/process/image, POST /api/process/complete for full multimodal fusion.
backend/routes/chat_routes.py	Python	Endpoint: POST /api/chat/ connecting front-end chat dialogue to Groq LLaMA-3.3.
backend/routes/results_bp.py	Python	Endpoints: GET /api/reports/download/ and GET /api/health for system diagnostics.
backend/services/vision_service.py	Python	Executes RT-DETR object detection, CLIP classification, depth map estimation, and 3D projection.
backend/services/audio_service.py	Python	Orchestrates Whisper ASR transcription, crime classification, and 4-language Groq translation.
backend/services/video_service.py	Python	Samples frames at 1Hz, orchestrates OpenCV computer vision engine, classification, second-by-second timeline.
backend/video_module/opencv_detector.py	Python	OpenCV frame differencing, contour analysis, people/vehicle/object detection, 3D coordinates, and timeline generation.
backend/video_module/api.py	Python	Endpoints: POST /api/video/opencv-predict and POST /api/analyze-video for OpenCV video intelligence.
backend/services/fusion_service.py	Python	Cross-modal entity resolution, consistency checking (audio vs video facts), master dossier.
backend/services/session_store.py	Python	Dual-tier memory and JSON disk persistence ensuring zero data loss upon server restart.
backend/services/pdf_service.py	Python	ReportLab Platypus PDF generator with XML escape sanitation, evidence catalog, and SHA-256 seal.

frontend/js/app.js	JavaScript	Main client controller; coordinates navigation tabs, processing trigger, view switching, PDF downloads.
frontend/js/scene_3d.js	JavaScript	Three.js WebGL canvas engine; renders room wireframe, camera OrbitControls, 3D evidence cubes.
frontend/js/timeline.js	JavaScript	Dynamic chronological timeline table rendering audio keypoints and 1.0 Hz video event cards.
frontend/js/chat_handler.js	JavaScript	Asynchronous chatbot dialogue manager with typing indicators, auto-scroll, and HTML sanitization.
frontend/js/api.js	JavaScript	Unified fetch API client wrapper managing all backend REST communications and error handling.